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Project #162: Nurse Driven NSTEMI Protocol

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Abstract

Introduction: NSTEMI patients are at risk for worsening ischemia, arrhythmia, and death when monitoring is inconsistent. At Henry Ford West Bloomfield Hospital, no standardized nurse driven protocol exists for serial troponins, ECG monitoring, or PRN antianginal management. This gap contributes to variability in care and delayed intervention. Evidence-based guidelines (ACC/AHA) support frequent troponin and ECG surveillance. By **December 31, 2025**, 95% of NSTEMI patients admitted to telemetry will begin receiving care aligned with this protocol. By **December 31, 2027**, the goal is that 90% of newly diagnosed NSTEMI patients will receive fully standardized monitoring, including EHR embedded charting.

Methods: The nurse driven protocol was developed to guide care with serial troponin measurements every three hours until values trend downward, daily ECGs with additional ECGs for ACS symptoms or rhythm changes, and PRN nitroglycerin management coordinated with providers and pharmacists. The protocol will be embedded in the EHR and reinforced through staff training with compliance and outcomes tracked through chart audits, workflow evaluation, and process measures.

Results: Preliminary audits will demonstrate improved compliance with troponin and ECG monitoring, faster recognition of deterioration, and stronger nurse–provider communication

Conclusions: This protocol improves patient safety, reduces variability, and strengthens nursing autonomy while showing promise for long-term sustainability and broader adoption across additional care settings

Plan

Problem Description

- NSTEMI patients face risk of ischemia, arrhythmia, and death with inconsistent monitoring.
- No standardized nurse driven protocol at HFWB for:
 - Serial troponins until downward trend
 - Daily ECG monitoring
 - PRN antianginal management
- Result: variability in care and delayed recognition of deterioration

Available Knowledge

- ACC/AHA guidelines:** recommend serial troponins and ECG surveillance.
- Evidence: early detection of myocardial injury improves outcomes.
- Literature: current practice is inconsistent, leading to variation and increased risk

Rationale

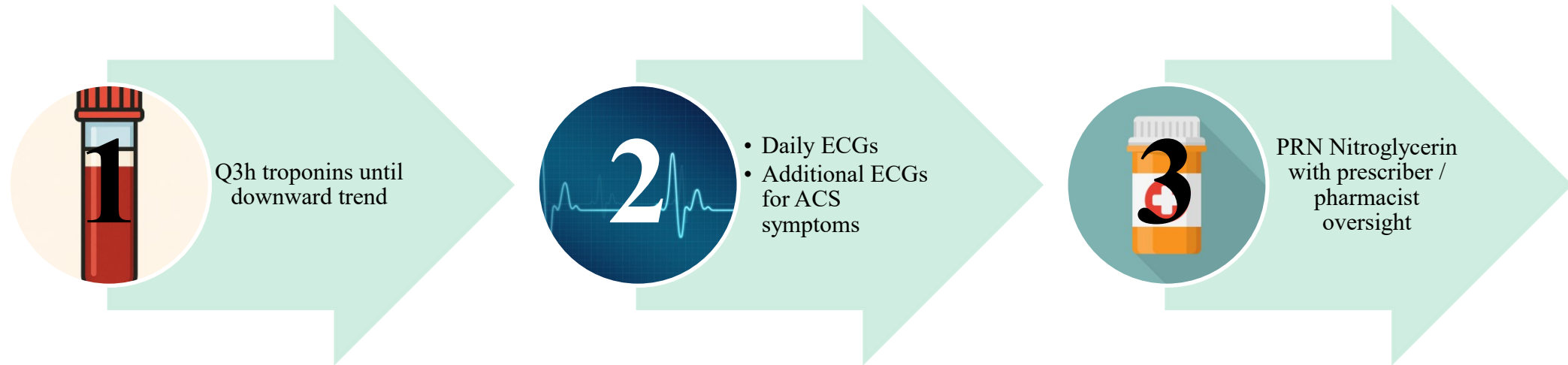
- Standardized nurse driven protocol reduces variability and aligns with evidence-based practice
- EHR integration, staff education, and workflow adaptation support improved compliance & patient safety
- Frameworks such as IHI Model for Improvement and PDSA cycles will guide phased rollout and sustainability
- Builds on existing nurse driven protocols at HFWB (e.g., Heparin and Sepsis), demonstrating feasibility, sustainability, and cultural readiness for expanding nurse-led initiatives

Specific Aim / Goals

Population: Context:	- Newly diagnosed NSTEMI patients (cardiac med-surg & stepdown); excludes CKD and Stage 2 MI - Added workload, additional lab draws and ECG demand
Timeline:	Phase 1 (2025) – Baseline audit, staff survey, stakeholder engagement, education, rollout Phase 2 (2026) – Ongoing monitoring, feedback, adjustments Phase 3 (2027) – Final analysis, outcome reporting, sustainability planning
Targets:	95% by Dec 2025: feasible early benchmark given available resources and expected adoption curve 90% by Dec 2027: sustainable long-term goal reflecting realistic resource capacity and workflow balance

Do

Intervention(s): Developed a nurse driven NSTEMI protocol including:



Study of the Intervention(s):

- EHR chart audits will compare outcomes *before vs after* protocol rollout
 - Was the protocol used for new NSTEMI patients?
 - Were extra ECGs obtained when indicated?
 - Was antianginal therapy appropriately given?
- Staff surveys & interviews will assess expected workflow changes and barriers
- Combined audit data & staff feedback to verify outcomes were due to the protocol, not random variation

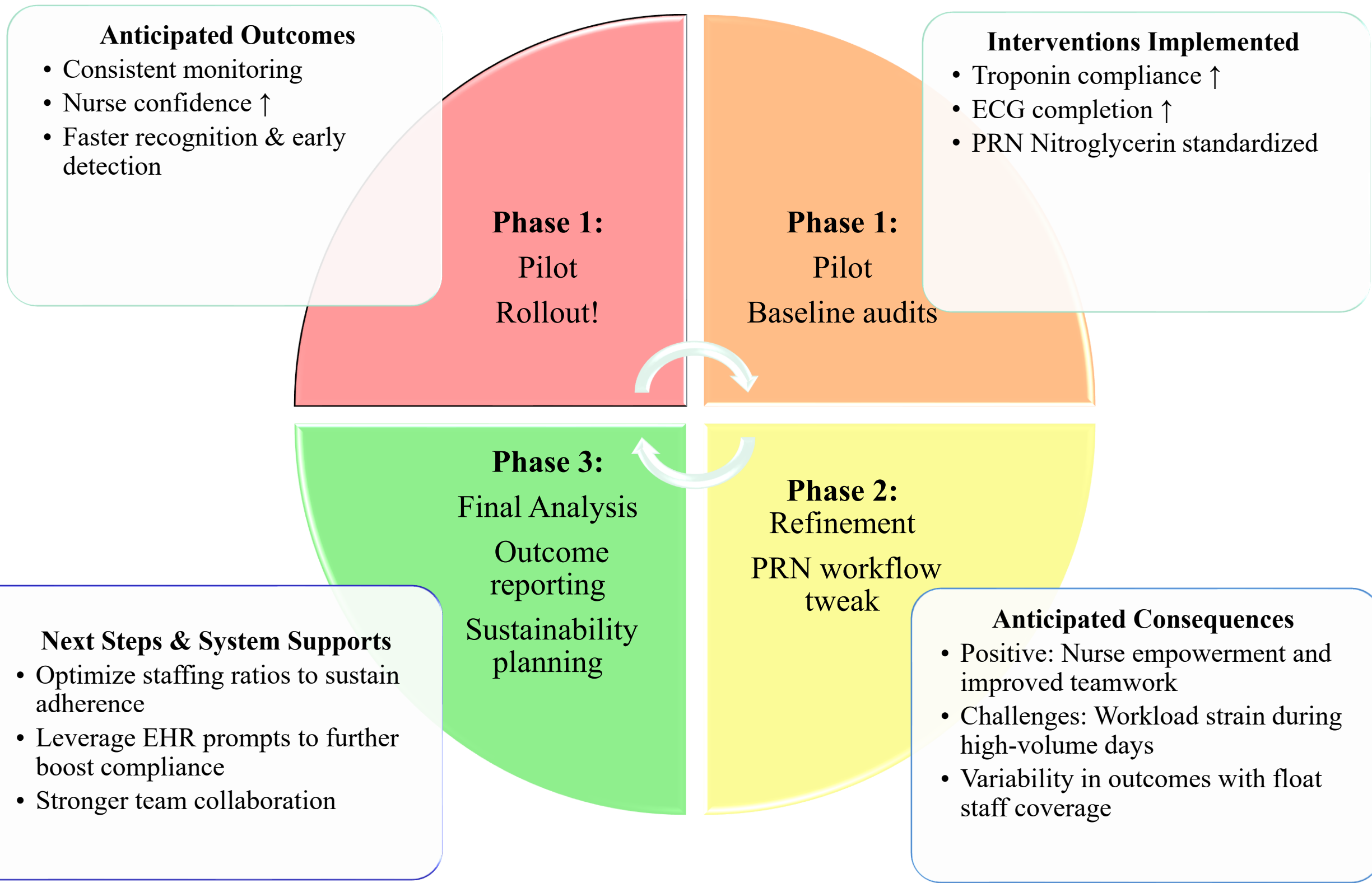
Measures:

- Process: Percentage of compliance with Q3h troponins, daily ECG completion, PRN medication documentation
- Outcome: timeliness of physician notification, early recognition of deterioration, transfers to higher care, 30-day rehospitalizations
- Balancing: staff workload, documentation burden, patient satisfaction
- Data accuracy will be checked through random chart reviews and EHR reporting validation

Analysis:

- Quantitative: run charts, compliance rates, outcome trends over time
- Qualitative: staff feedback analyzed to identify barriers/facilitators
- Mixed-methods analysis to confirm improvements linked to intervention

Check



Act

Summary

- Plan to evaluate compliance with troponin draws, ECG monitoring, and PRN nitroglycerin use after protocol rollout
- Strengths: EHR integration, structured staff education, and workflow tools that support nursing autonomy

Interpretation

- Anticipated outcomes: faster recognition of deterioration and improved nurse–provider communication
- Results will be compared with ACC/AHA guidelines and other nurse driven QI projects.
- Impact expected on patient safety, interdisciplinary teamwork, and timely escalation of care.
- Costs/Trade-offs:** increased frequency of labs and ECGs may add to staff workload and resource use but are expected to reduce downstream costs through earlier detection and fewer adverse events
- Quantitative results statement:** full compliance and outcome data will be presented after final analysis

Limitations

- Generalizability is limited to telemetry units, specifically the cardiac medical-surgical and stepdown units. Patients with chronic kidney disease and those presenting with stage 2 MI are excluded.
- Potential threats to validity: staffing variation, patient acuity, and documentation practices
- To mitigate these limitations, we will use standardized audit processes, consistent data collection, and feedback loops embedded in PDSA cycles

Conclusions

- Nurse driven NSTEMI protocol has the potential to improve patient safety, reduce care variability, and strengthen nursing autonomy
- Sustainability supported through EHR integration and ongoing staff training
- Builds on established nurse driven protocols at HFWB (e.g., Heparin, Sepsis), showing organizational readiness for long-term adoption
- Next steps: Begin phase one, then complete data analysis, track patient-centered outcomes (i.e., adverse events, mortality), and explore spread to additional care settings

Other Information

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Ethical Considerations: *This project represents a proposed quality improvement initiative. No patient-level data has been collected and no intervention is currently in effect. Should the protocol move forward, ethical considerations, including institutional review board (IRB) determination and balancing of staff workload with patient benefit, will be addressed in alignment with Henry Ford Health policies. No funding or conflicts of interest to disclose.*

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